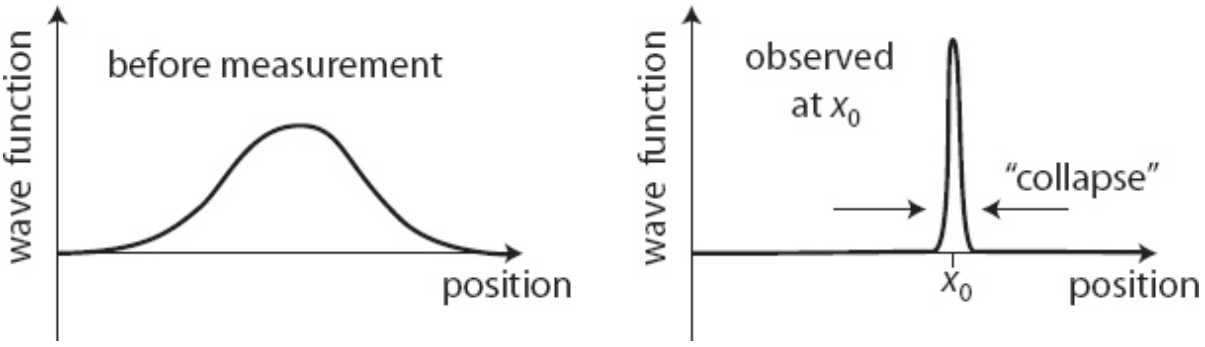
predict which of the results it will be, but you can calculate the probability for each allowed outcome. And after your measurement is done, the wave function *collapses* to a completely different function, with all of the new probability concentrated on whatever result you just got. So if you measure a quantum system, in general the best you can do is predict probabilities for various outcomes, but if you were to immediately measure the same quantity again, you will always get the same answer—the wave function has collapsed onto that outcome.

Let's write this out in gory detail.

Rules of Quantum Mechanics (Part Two)

3. There are certain observable quantities we can choose to measure, such as position, and when we do measure them, we obtain definite results.
4. The probability of getting any one particular result can be calculated from the wave function. The wave function associates an amplitude with every possible measurement outcome; the probability for any outcome is the square of that amplitude.
5. Upon measurement, the wave function collapses. However spread out it may have been pre-measurement, afterward it is concentrated on the result we obtained.



In a modern university curriculum, when physics students are first exposed to quantum mechanics, they are taught some version of these five rules. The ideology associated with this presentation—treat measurements as fundamental, wave functions collapse when they are observed, don't ask questions about what's going on behind the scenes—is sometimes called the